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Penetration and accumulation of ions into living cells or tissues from surrounding medium by crossing membrane is called **mineral absorption**. Movement of ions into and out of cells or tissues is termed as transport or **flux**. Entry of the ion into cell is called **influx** and exit is called **efflux**. Various theories have been put forward to explain this mechanism. They are categorized under passive mechanisms (without the involvement of metabolic energy) and active mechanisms (involvement of metabolic energy).

### 11.8.1 Passive Absorption

**1. Ion-Exchange:**

Ions of external soil solution were exchanged with same charged (anion for anion or cation for cation) ions of the root cells. There are two theories explaining this process of ion exchange namely:

i. Contact exchange and ii. Carbonic acid exchange.

i. **Contact Exchange Theory:**

According to this theory, the ions adsorbed on the surface of root cells and clay particles (or clay micelles) are not held tightly but oscillate within a small volume of space called **oscillation volume**. Due to small space, both ions overlap each other's oscillation volume and exchange takes place (Figure 11.23).

ii. **Carbonic Acid Exchange Theory:**

According to this theory, soil solution plays an important role by acting as a medium for ion exchange. The $CO_2$ released during respiration of root cells combines with water to form carbonic acid ($H_2CO_3$). Carbonic acid dissociates into $H^+$ and $HCO_3^-$ in the soil solution. These $H^+$ ions exchange with cations adsorbed on clay particles and the cations from micelles get released into soil solution and gets adsorbed on root cells (Figure 11.24).

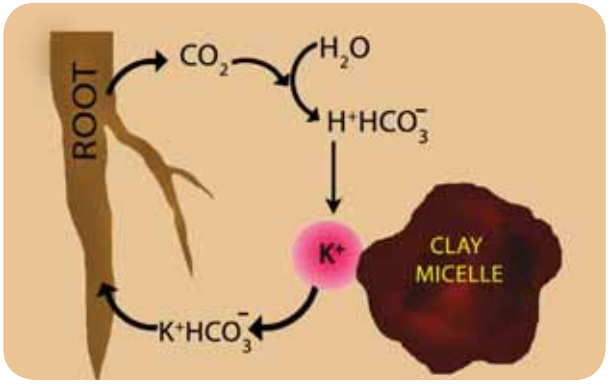


**Figure 11.24:** Carbonic Acid Exchange theory

**Figure 11.23:** Contact Exchange theory